

Module A

Gu, Kelly, Xiu (2020)

Barbara Będowska-Sójka · Poznań University of Economics and Business

MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



UNIWERSYTET
EKONOMICZNY
W POZNANIU

Papers and materials

Gu, S., Kelly, B., & Xiu, D. (2020).
Empirical Asset Pricing via Machine Learning.
Review of Financial Studies, 33(5), 2223–2273.

After 01 (FF + lagged ML).

Live: notebooks/01a_gu_kelly_xiu_simplified.ipynb

Authors' data & code: datashare.zip (Xiu's website) ·

github.com/xiubooth/ML_Codes

Paper card: Gu, Kelly & Xiu (2020)

	<i>Review of Financial Studies</i> 33(5): 2223–2273
Question	Which ML methods measure the equity risk premium best, by predicting individual stock returns one month ahead?
Data	About 30,000 US stocks (CRSP), March 1957 – December 2016; 94 characteristics interacted with 8 macro predictors, plus 74 industry dummies: 920 predictors.
Method	A ladder from OLS and elastic net through PCR/PLS and generalized linear models to random forests, boosted trees and neural nets with 1–5 hidden layers.
How it is tested	Train 1957–74, validate 1975–86, test 1987–2016, refit every year; no cross-validation, to keep time order. Out-of-sample R^2 against a zero forecast; long–short decile portfolios.
Main result	Trees and neural nets beat linear models; shallow nets do best. The best monthly R^2_{OOS} is small (→ poll in class). Long–short Sharpe ratio of the neural net: 1.35 value-weighted, 2.45 equal-weighted.
Watch for	The sample keeps stocks below \$5. Equal-weighted beats value-weighted: where does the profit come from? (→ Avramov et al.)
Code & data	Characteristics (datashare.zip) and code (GitHub: xiubooth/ML_Codes) are public; returns need CRSP.

Economic question

Measure the equity **risk premium** by predicting individual stock excess returns one month ahead.

Which ML tools beat linear benchmarks **out of sample**, in forecast accuracy and in portfolios?

Hypothesis: nonlinear interactions among characteristics carry risk-premium information that linear models miss.

Target & sample (paper)

- Unit: stock–month, all NYSE/AMEX/NASDAQ stocks in CRSP
- March 1957 – December 2016: almost 30,000 stocks, more than 6,200 per month
- **Kept:** stocks priced below \$5, non-common share codes, financials (fn. 28)
- Target: next-month excess return (T-bill as the risk-free rate)

Classroom notebook: a synthetic panel with a planted nonlinear signal.

Features: 920 predictors

Block	Count
Firm characteristics (61 annual, 13 quarterly, 20 monthly)	94
× Welch–Goyal macro (dp, ep, bm, ntis, tbl, tms, dfy, svar)	8
Industry dummies (2-digit SIC)	74

Characteristics are **ranked** cross-sectionally each month into $[-1, 1]$.

Ask: what is known **before** the month $t + 1$ return?

Validation

- Training 1957–1974 (18 y) · validation 1975–1986 (12 y) · **test 1987–2016 (30 y)**
- Refit **once a year**: the training window expands, the 12-year validation window rolls forward
- **No** cross-validation, "to maintain the temporal ordering of the data"
- Tune on validation; the test period stays untouched

Results: small R^2 , clear ranking

Model	R^2_{OOS} (monthly, %)
OLS, all 920 features	−3.46
OLS-3 (size, bm, mom)	0.16
ENet + Huber	0.11
RF / GBRT + Huber	0.33 / 0.34
NN1 → NN3	0.33 → 0.40

R^2_{OOS} is measured against a forecast of **zero**, not the historical mean.

Finance evaluation

Long–short decile spread on NN forecasts:

annualized Sharpe **1.35** (VW) / **2.45** (EW)

against 0.61 / 0.83 for the OLS benchmark.

The dominant signals are momentum, liquidity and volatility.

Discuss: EW beats VW by a lot. What does that tell you?

Model ladder (paper → classroom)

Paper	Classroom subset
OLS / PCR / PLS / elastic net	OLS, Ridge
Trees / RF / GBRT	Random Forest
Neural nets	Small MLP

Always keep the linear baseline.

Homework: 01b (named characteristics).

Live notebook agenda (~45 min)

1. Build the synthetic characteristic panel
2. Temporal split
3. Fit the ladder
4. OOS R^2 table
5. Long–short cumulative paths
6. One "do differently" cell

Do differently (seeds)

1. Rank-transform on/off
2. Expanding vs rolling window
3. Turnover penalty
4. **Drop microcaps** (below the 20th NYSE size percentile) and re-sort

The last seed is the next paper.

Paper card: Avramov, Cheng & Metzker (2023)

	<i>Management Science</i> 69(5): 2587–2619
Question	Do deep-learning return signals survive economic restrictions: microcaps, distressed firms, trading costs?
Data	US stocks, out-of-sample 1987–2017; signals from GKX's NN3, CPZ's GAN, IPCA and the conditional autoencoder.
Method	Re-sort the ML signals after excluding microcaps (below the 20th NYSE size percentile), distressed firms (12 months around a rating downgrade) and high limits-to-arbitrage periods.
How it is tested	Value-weighted, FF6-adjusted long–short returns; portfolio turnover.
Main result	GKX's FF6-adjusted return falls from 0.92% to 0.31% a month without microcaps. Without distressed firms, no deep-learning method is significant. Turnover is at least 87% a month. The signals still pay on the long side and in crashes.
Watch for	Different filters, benchmark and weighting from GKX. Which claim is each paper testing?
Code & data	Needs CRSP/Compustat and the original authors' signals.

Robustness: who earns Gu's alpha?

Avramov, D., Cheng, S., & Metzker, L. (2023). *Machine Learning vs. Economic Restrictions: Evidence from Stock Return Predictability*. *Management Science*, 69(5), 2587–2619.

- Models: GKX NN3, CPZ GAN, IPCA, conditional autoencoder
- Out-of-sample 1987–2017
- **Microcaps**: below the 20th NYSE size percentile
- **Distressed**: rated firms from 12 months before to 12 months after a downgrade

Robustness: the numbers

	GKX, VW, FF6-adjusted
All stocks	$\approx 0.92\%$ / month
Without microcaps	$\approx 0.31\%$ / month ($\approx -2/3$)
Without distressed firms	not significant

"None of the deep learning methods generates significant value-weighted FF6-adjusted returns at the 5% level after excluding distressed firms."

GKX turnover: $\geq 87\%$ per month.

On the other side: the long leg is profitable, recent years are profitable, and downside risk is low.

Discussion: both papers can be right

GKX (fn. 28): results unchanged when low-price stocks, non-common shares and financials are filtered out.

Avramov et al.: excluding microcaps or distressed firms "considerably attenuates profitability".

What differs?

- the **filter** (price and share code vs size, rating and volatility state)
- the **benchmark** (raw return vs FF6 alpha)
- the **weighting** (EW vs VW)

Which hypothesis does each paper test?

Takeaway

Gu–Kelly–Xiu is a **research design** for ML in the cross-section, not a verdict that "neural nets win".

Where the alpha lives is part of the result.

Next: Krauss, with classification and a portfolio rule